SAMRATASHOKTECHNOLOGICALINSTITUTE
(Engineering College),VIDISHA M.P.
(An Autonomous Institute Affiliated to RGPV Bhopal)
Department of Artificial Intelligence

Semester/Year			Program		B.Tech.
Subject Category	DLC	Subject Code:	DS26101P	Subject Name:	**Computer Workshop**

Maximum Marks Allotted						Contact Hours			Total Credits
Theory			Practical		TotalMarks				
End Sem	Mid-Sem	Assignment / Quiz	End Sem	Lab- Work		L	T	P	
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Course Objective:

1. To teach principles of operating system including File handling utilities, Security by file permissions, Process utilities, Disk utilities, Networking Commands, Basic Linux commands, Scripts and filters.
2. To familiarize fundamentals of the Bourne again shell (bash), shell programming, pipes, input and output redirection Control structures, arithmetic in shell interrupt processing, functions, debugging shell scripts.
3. To impart fundamentals of file concepts kernelsupport for file, File structure related systemcalls (file API's).
4. TofacilitatestudentsinunderstandingInterprocesscommunication.
5. Tofacilitatestudentsinunderstandingsemaphoreandsharedmemory.
6. Tofacilitatestudentsinunderstandingprocess.

Course Outcomes:

Upon completion of this course ,the student will be able to:

CO1.AbilitytousevariousLinuxcommandsthatareusedtomanipulatesystem operations at admin level and a prerequisite to pursue job as a Network administrator.

CO2.AbilitytowriteShellProgrammingusingLinuxcommands.

CO3. Ability to design andwrite application to manipulate internal kernel level Linux FileSystem.

CO4.AbilitytodevelopIPC-API'sthatcanbeusedtocontrolvariousprocessesfor synchronization.

CO5.AbilitytodevelopNetworkProgrammingthatallowsapplicationstomakeefficient useofresourcesavailableondifferentmachinesinanetwork.

UNITS	Descriptions	Hrs.	CO's
I	**INTRODUCTIONTOLINUXANDLINUXUTILITIES**:Abriefhistory of LINUX, architecture of LINUX, features of LINUX, introduction tovi editor. Linux commands, file handling utilities, security by file permissions, process utilities, disk utilities, networking commands, Text Processing utilities and backup utilities	4	CO1
II	**Introduction to Shells**: Linux Session, Standard Streams, Redirection, Pipes, Command-Line Editing, Options, Shell/Environment Customization. **Filters:** Filters and Pipes ,Concatenating files, Display Beginning and End of files, Operations on Characters, Files with Duplicate Lines, Count Characters, Words or Lines, Comparing Files.	4	CO2
III	**Grep:**Operation, grep Family, Searching for File Content. **Sed**: Scripts, Operation, Addresses, commands, Applications, grep and sed. UNIX FILE STRUCTURE: Introduction to UNIX file system, inode (Index Node), file descriptors, system calls and device drivers. **File Management**: File Structures, System Calls for File Management,DirectoryAPI.	4	CO3
IV	**PROCESS AND SIGNALS**: Process, process identifiers, process structure: process table, viewing processes, system processes, processscheduling,zombieprocesses,orphanprocess,unreliable signals,interruptedsystemcalls. **Filelocking**:creatinglockfiles,lockingregions,useofreadandwrite with locking, competing locks, other lock commands, deadlocks.	4	CO4
V	**INTERPROCESSCOMMUNICATION**: Pipe, process pipes,thepipe call, parent and child processes, and named pipes, semaphores, message queues, shared memory. **INTRODUCTION TO SOCKETS**: Socket, socket connections - socket attributes, socket addresses.	4	CO5

GuestLectures(ifany)	--	
TotalHours	20	

ListofExperiments

1. Writeaprogramusingecho,printf,script,passwd,uname,who,date,stty,pwd commands.
2. Write a program using unlink, du, df,mount, umount, find, unmask, ulimit, ps, w, finger,arp, ftp commands.
3. Writeaprogramusingtelnet,rlogin.TextProcessingutilitiesandbackuputilities,tail,head ,sort,nl,uniq,grep,egrep,fgrep,cut,paste,join,tee,pg,comm,cmp,diff,tr,awk commands.
4. Writeashellscriptthatacceptsafilename,startingandendinglinenumbersasarguments and displays all the lines between the given line numbers.
5. Illustrate bywriting scriptthat will print, message "HelloWorld, inBold andBlink effect, and in different colours like red, brown etc using echo commands?
6. Write a shell script that deletes all lines containing a specified word in one ormore filessupplied as arguments to it.
7. Illustratebywritingscriptusingforlooptoprintthefollowingpatterns?
8. Write a shell script that displays alist ofall the filesin the currentdirectory towhich the user has read, write and execute permissions.
9. Writeaprograminter-processcommunication.
10. Writeaprogramtocommunicateusingsockets.

TextBooks-

1. W. Richard. Stevens (2005), Advanced Programming in the UNIX Environment, 3rd edition, Pearson Education, New Delhi, India.
2. UnixandshellProgrammingBehrouzA.Forouzan,RichardF.Gilberg.Thomson

REFERENCESBooks-:

1. LinuxSystemProgramming,RobertLove,O'Reilly,SPD.
2. AdvancedProgrammingintheUNIXenvironment,2ndEdition,W.R.Stevens,Pearson Education.
3. UNIXNetworkProgramming,W.R.Stevens,PHI.UNIXforProgrammersandUsers,3rd Edition,GrahamGlass,KingAbles,PearsonEducation

Modes of Evaluation and Rubric

The evaluation modes consist of performance in Quiz/ Assignments, term work, and end-semester practical examinations.

List/Links of e-learning resource

- https://onlinecourses.nptel.ac.in
- https://www.classcentral.com(swayam)

Recommendation by Board of studies on	June-2026
Approval by Academic council on	June-2026
Compiled and designed by	AI
Subject handled by department	AI